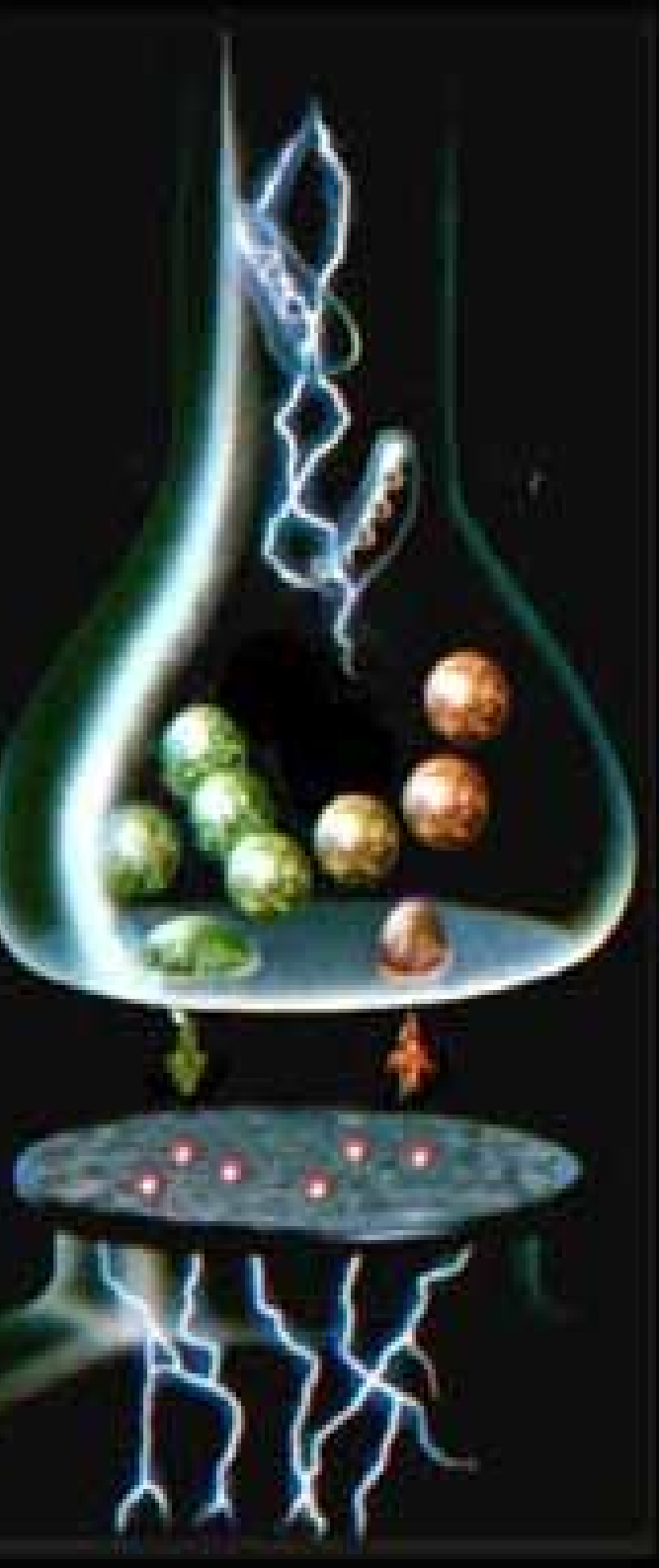


Synapse and Neuromuscular junction



Dr.Wafa Mubarak



Objectives

- *Define synapse*
- *Anatomy and structure of synapse*
- *Function of synapse*
- *Difference between chemical & electrical synapse*
- *Properties of synapse*
- *Neuromuscular junction*



Definition of synapse:

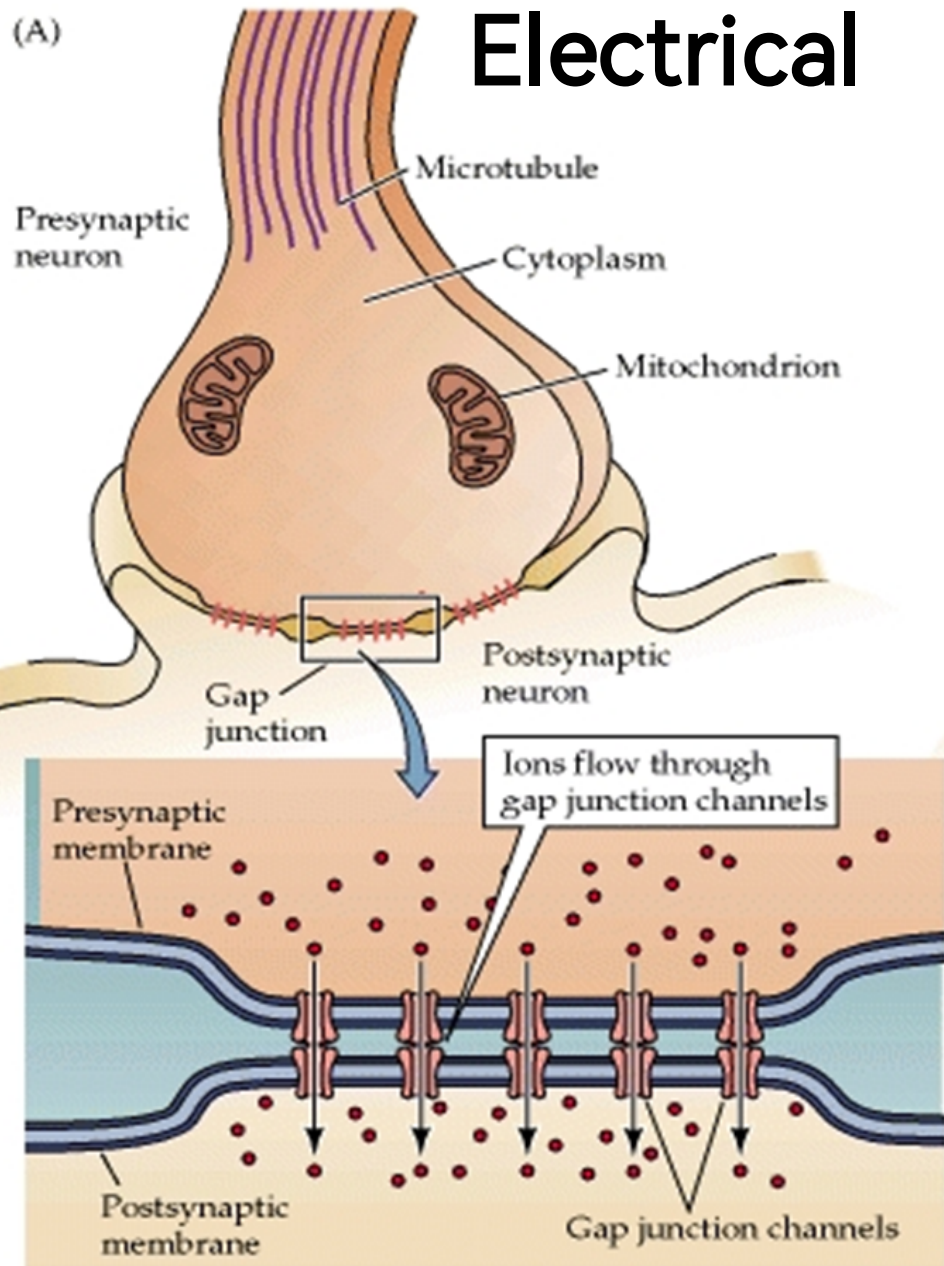
- *Junction between two neuron.*
- *Nerve impulse is an action potential propagated along the nerve of the neuron.*
- *Nerve impulse is transmit to other cell at the synapse*



Types of synapse

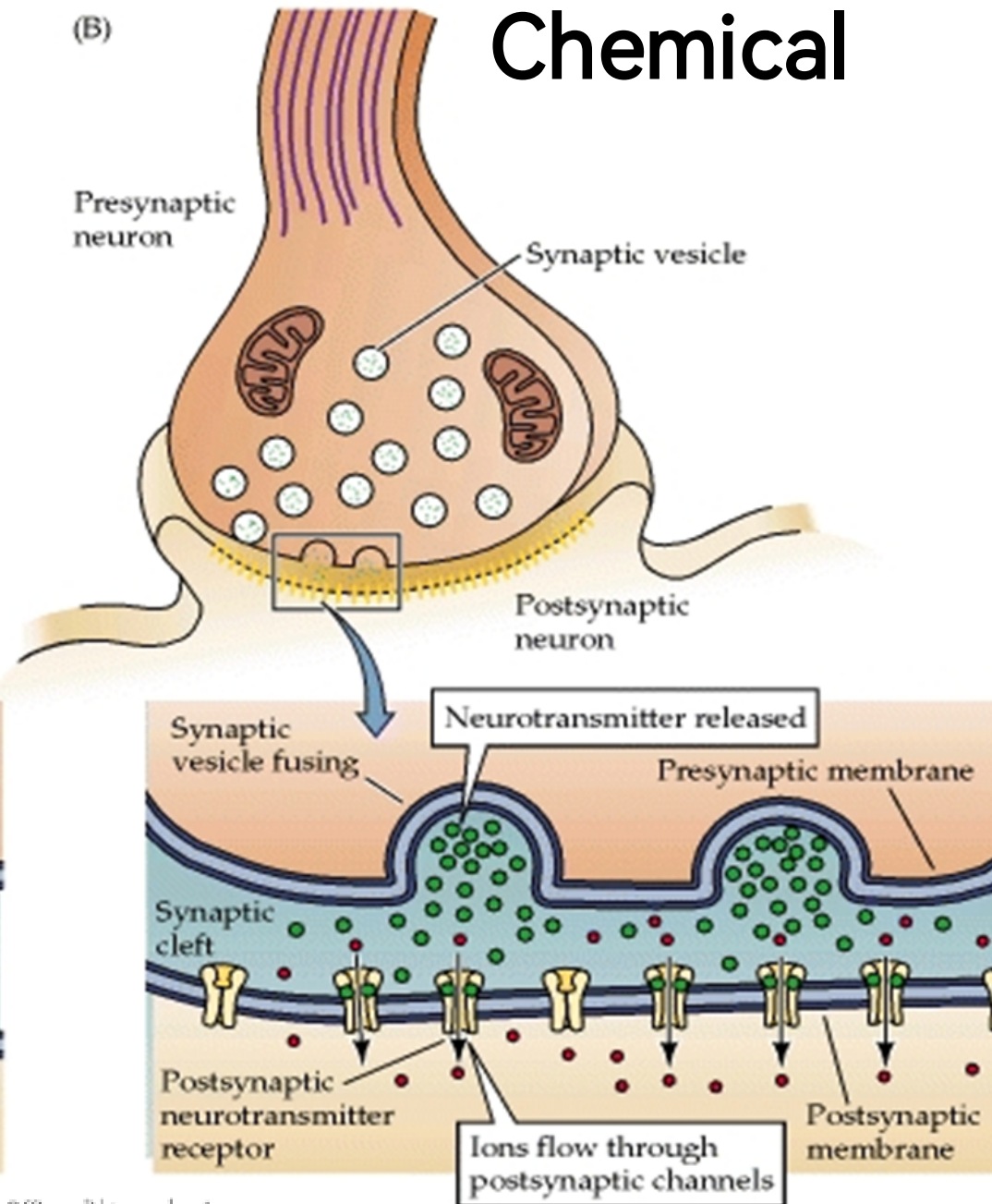
(A)

Electrical



(B)

Chemical



Electrical synapse

- In these synapses the membranes of the two cells connected, and they share proteins and electrolytes.
- This allows the action potential to pass directly from one membrane to the next.
- They are very fast, but are quite rare, found mainly in the heart .

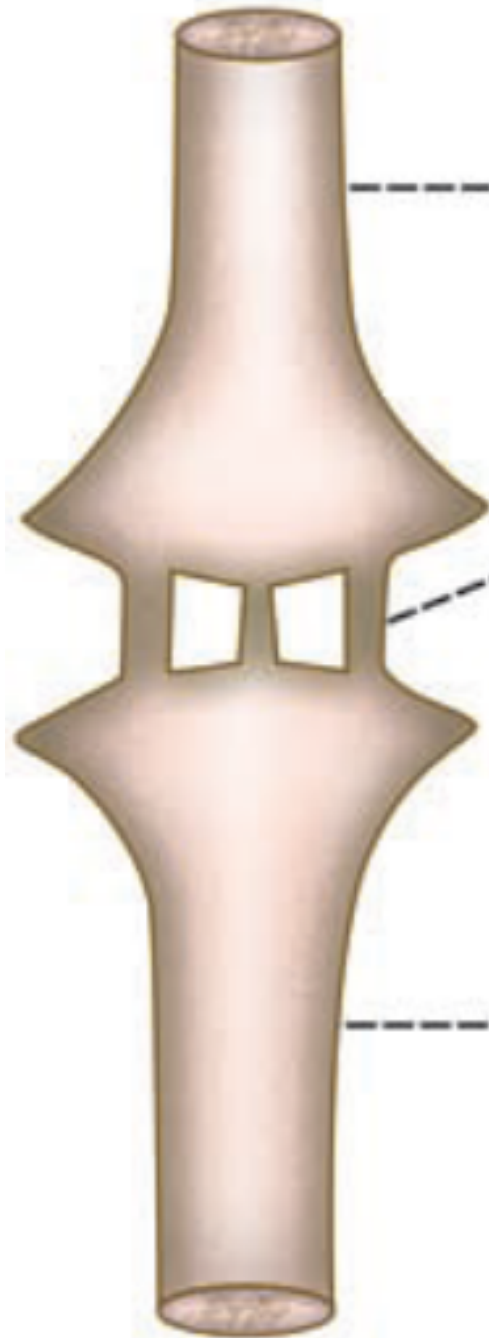


Chemical synapse

- In a chemical synapse, electrical activity in the presynaptic neuron is converted into the release of a chemical called a neurotransmitter that binds to receptors located in the plasma membrane of the postsynaptic cell.



Electrical synapse



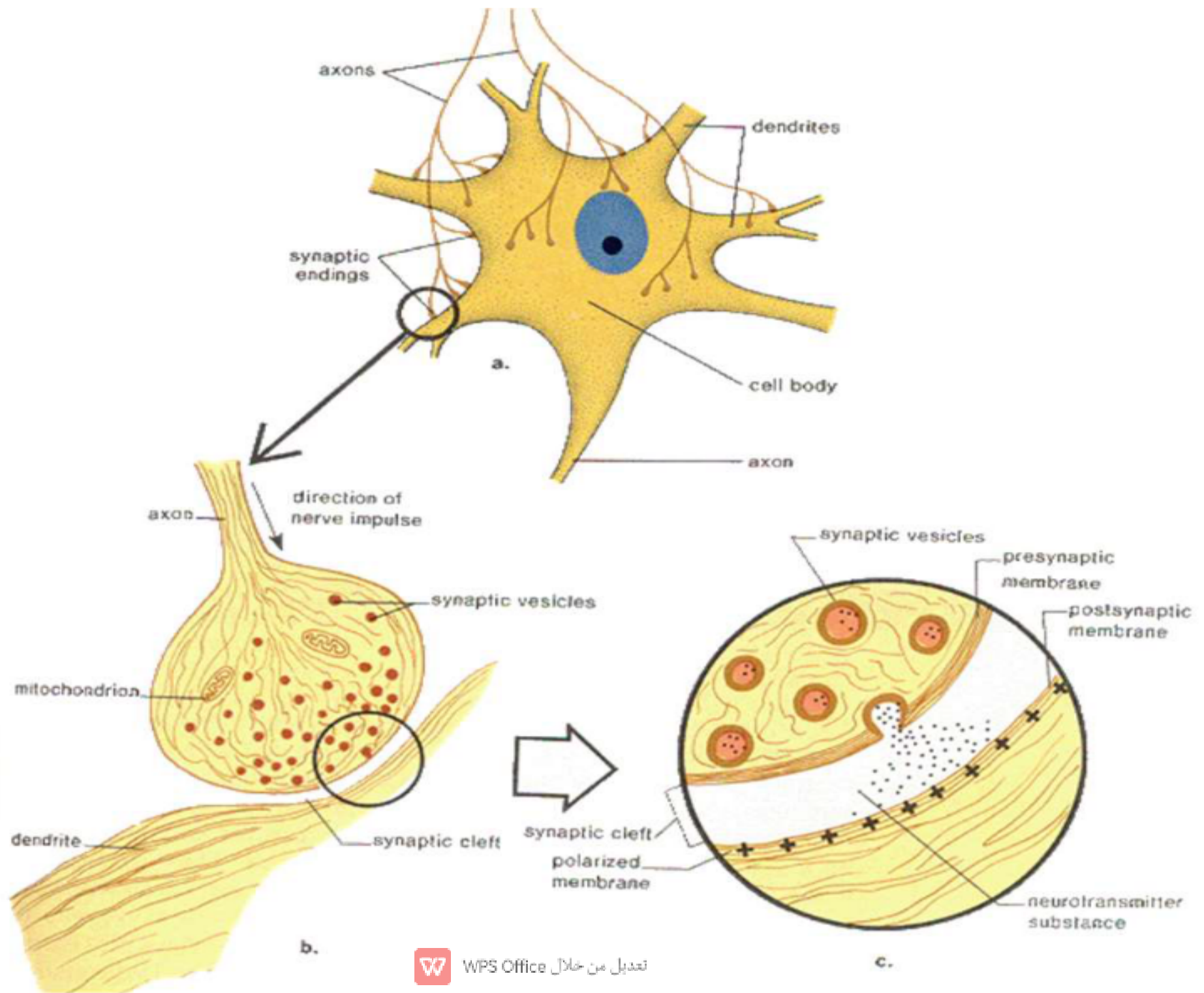
Chemical synapse

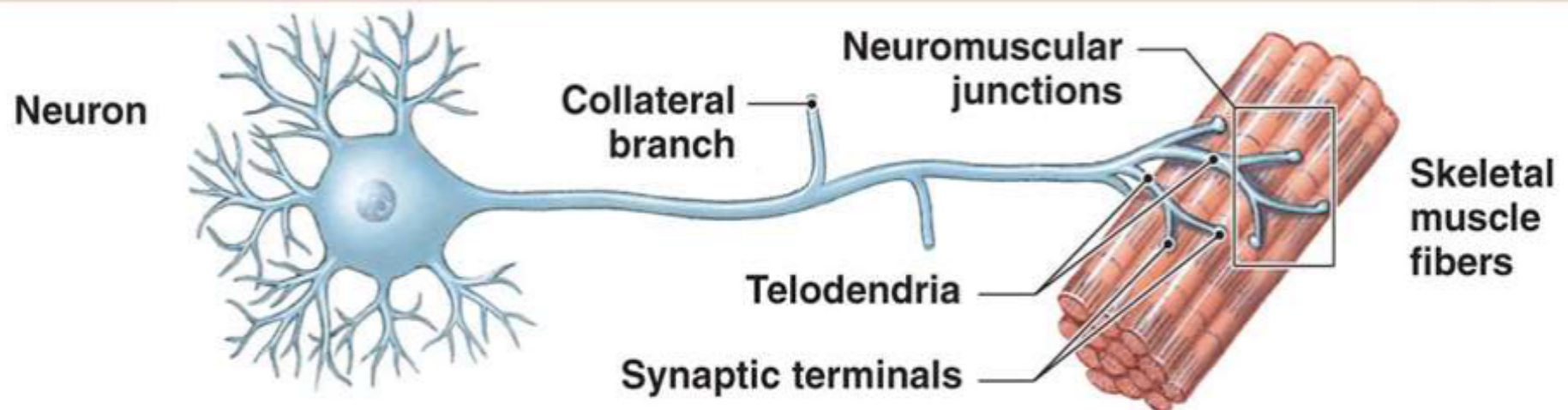
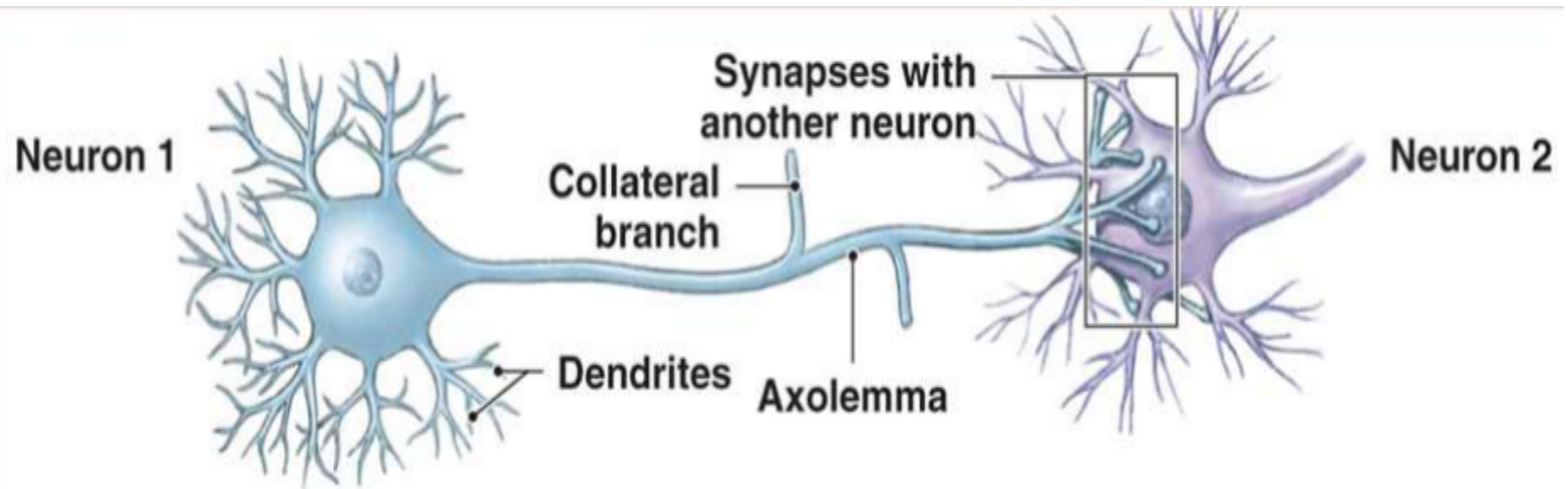


Structure of synapse

- *Presynaptic neuron : neuron send impulse*
- *Postsynaptic neuron : neuron receive impulse .*
- *Synaptic cleft : separate them (is 20—40nm wide) .*
- *Receptor in postsynaptic membrane .*
- *In presynaptic terminal : many mitochondria & membrane enclosed vesicles contain neurotransmitter .*







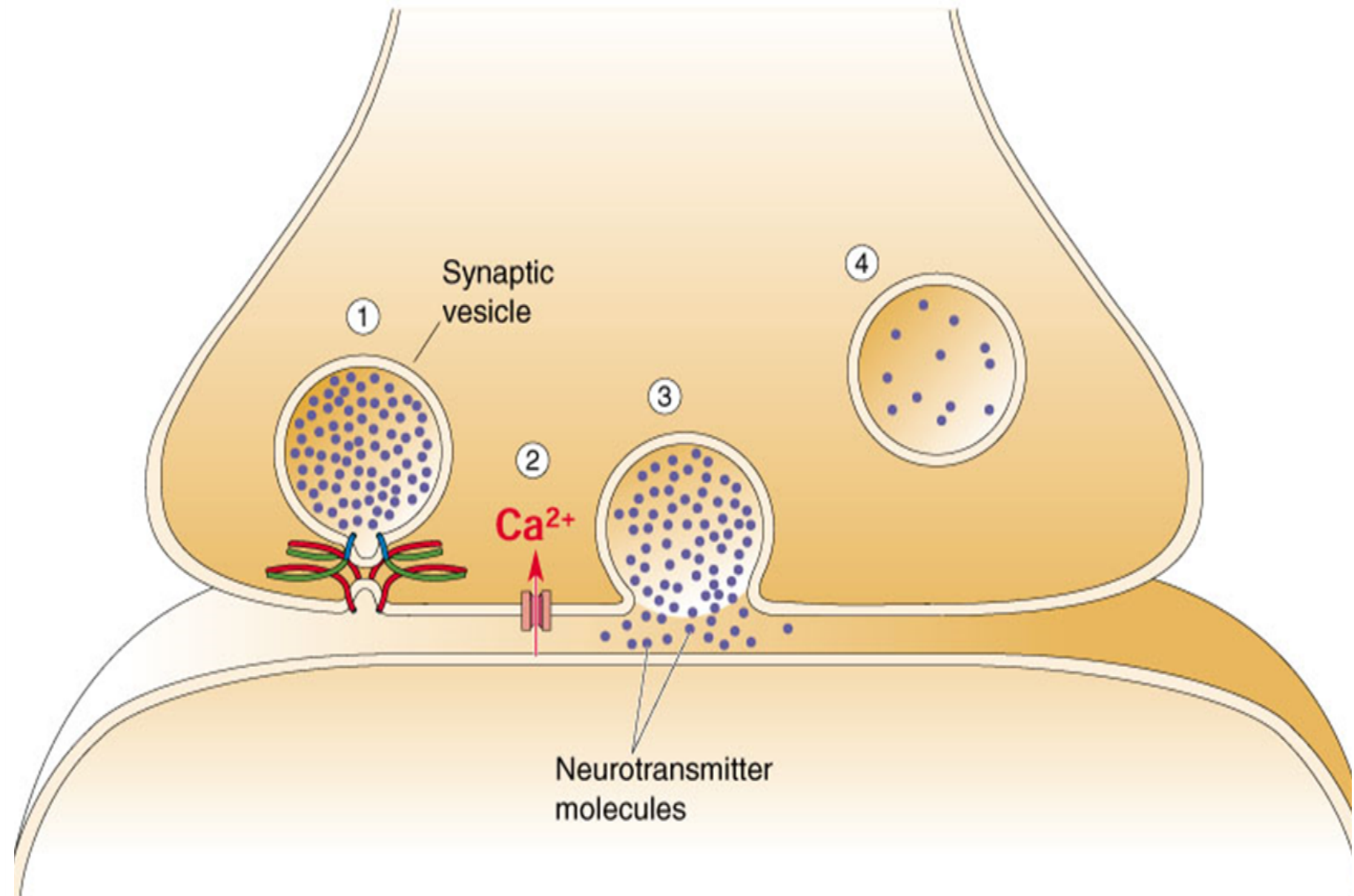
Transmission of neurotransmitters

- Fusion of the synaptic vesicles with the cell membrane
- releasing their contents (the neurotransmitter chemicals) by exocytosis.
- The neurotransmitters diffuse across the synaptic cleft.
- The neurotransmitter binds to the neuroreceptors in the post-synaptic membrane, causing the channels to open and transport signals.



Figure 5.10

The release of neurotransmitter by exocytosis. A synaptic vesicle loaded with neurotransmitter, in response to an influx of Ca^{2+} through voltage-gated calcium channels, 1 releases its contents into the synaptic cleft by the fusion of the vesicle membrane with the presynaptic membrane, and 2 is eventually recycled by the process of endocytosis.



FUNCTIONS OF SYNAPSE

- Main function of the synapse is to transmit the impulses, i.e. action potential from one neuron to another.
- However, some of the synapses inhibit these impulses. So the impulses are not transmitted to the postsynaptic neuron.



- On the basis of functions, synapses are divided into two types:
- 1. Excitatory synapses, which transmit the impulses (excitatory function)
- 2. Inhibitory synapses, which inhibit the transmission of impulses (inhibitory function).



Presynaptic neuron

Arrival of action potential in axon terminal

Opening of calcium channels in presynaptic membrane

Influx of calcium ions from ECF into the axon terminal

Opening of vesicles and release of Ach

Passage of Ach through synaptic cleft

Postsynaptic neuron

Formation of Ach-receptor complex

Opening of sodium channels and influx of sodium ions from ECF

Development of EPSP

Opening of sodium channels in initial segment of axon

Influx of sodium ions from ECF and development of action potential

Spread of action potential through axon of postsynaptic neuron



Presynaptic neuron

Arrival of action potential in axon terminal

Opening of calcium channels in presynaptic membrane

Influx of calcium ions from ECF into axon terminal

Opening of vesicles and release of GABA

Passage of GABA through synaptic cleft

Postsynaptic neuron

Formation of GABA-receptor complex

Opening of K^+ and efflux of K^+ into ECF

Opening of Cl^- and influx of Cl^- into neuron

More negativity inside postsynaptic neuron

Hyperpolarization – IPSP

Inhibition of synaptic transmission



Neurotransmitters

- a chemical substance that is released at the end of a nerve fiber, causes the transfer of the impulse to another nerve or muscle fiber.
- Excitatory neurotransmitters can cause the next cell to initiate a new impulse:
- Acetylcholine
- Norepinephrine
- Glutamate



- Inhibitory neurotransmitters they reduce chances of a new impulse:
- GABA
- Glycine



Synaptic properties

1. One-way conduction

*Synapses generally permit
conduction of impulses in one-
way*



2-Synaptic delay

- Is the minimum time required for transmission across the synapse.*
- it's about 0.2-0.5 ms*

3-Fatigue

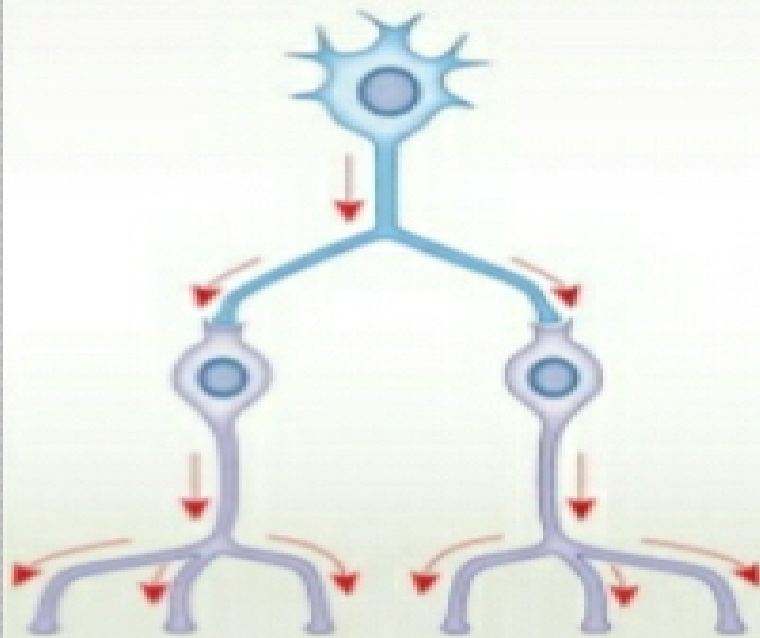
Exhaustion of nerve transmitter.



4- Convergence and divergence

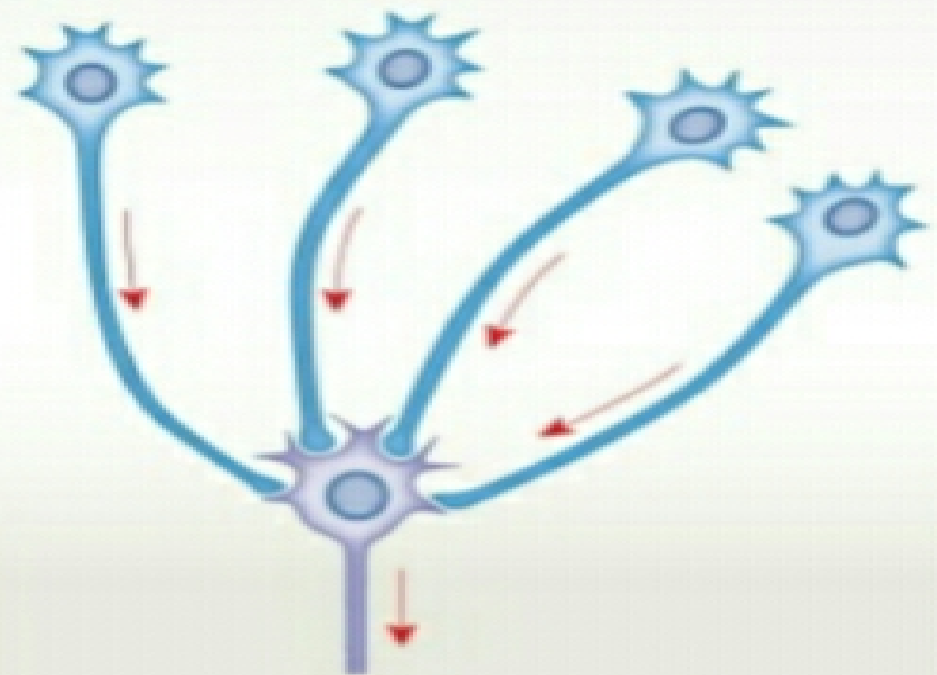
a Divergence

A mechanism for spreading stimulation to multiple neurons or neuronal pools in the CNS

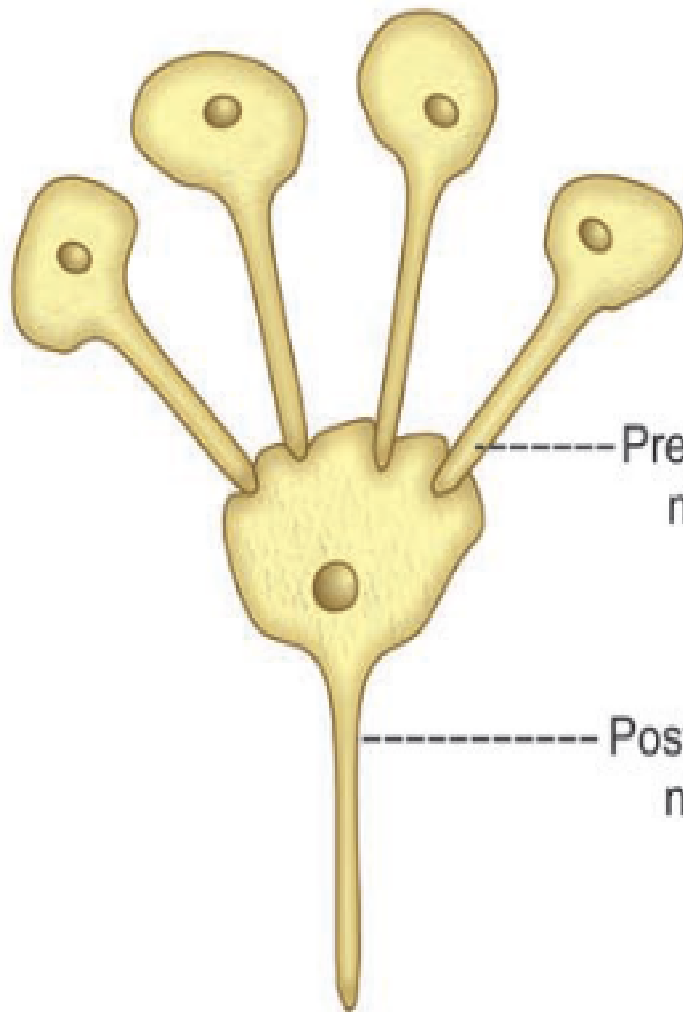


b Convergence

A mechanism for providing input to a single neuron from multiple sources



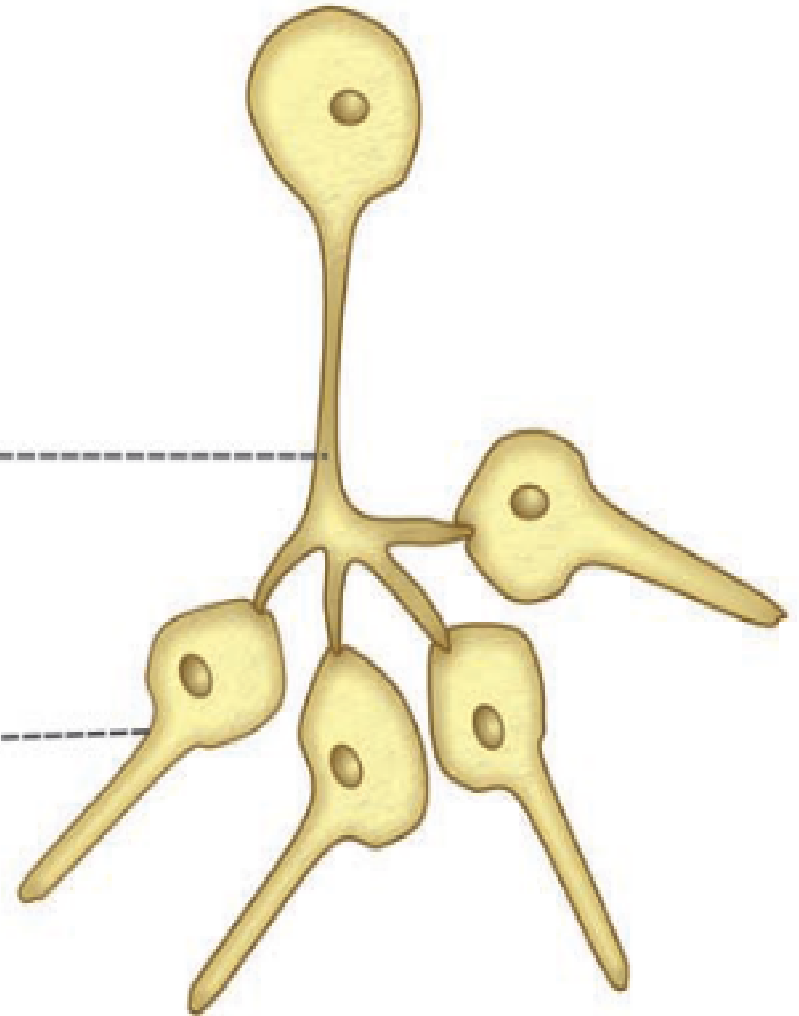
Convergence



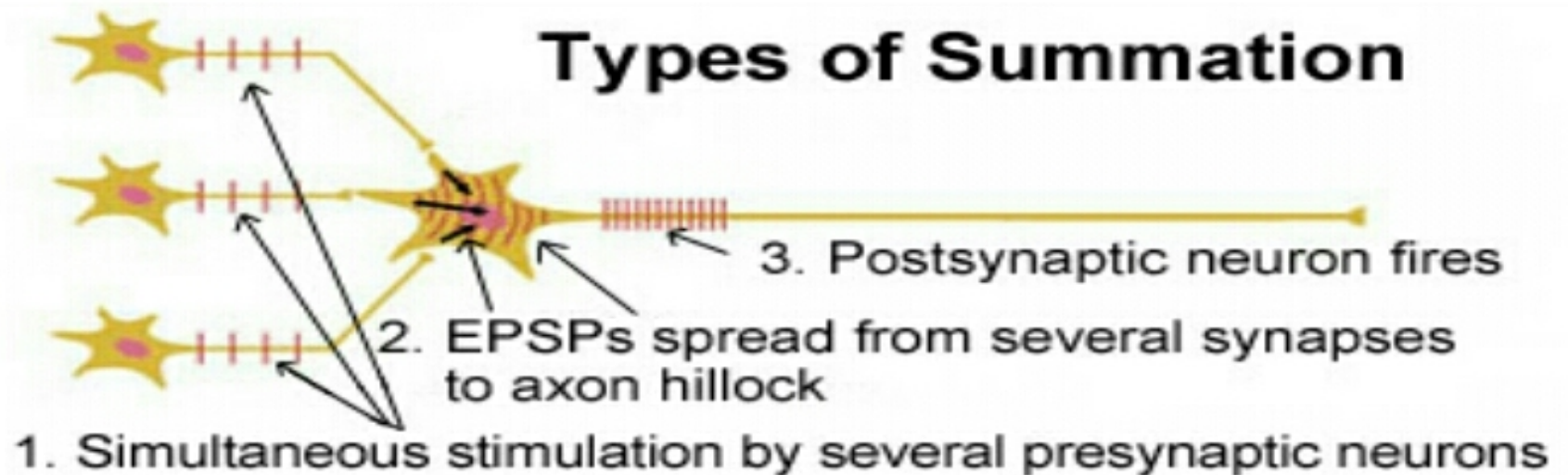
Presynaptic
neuron

Postsynaptic
neuron

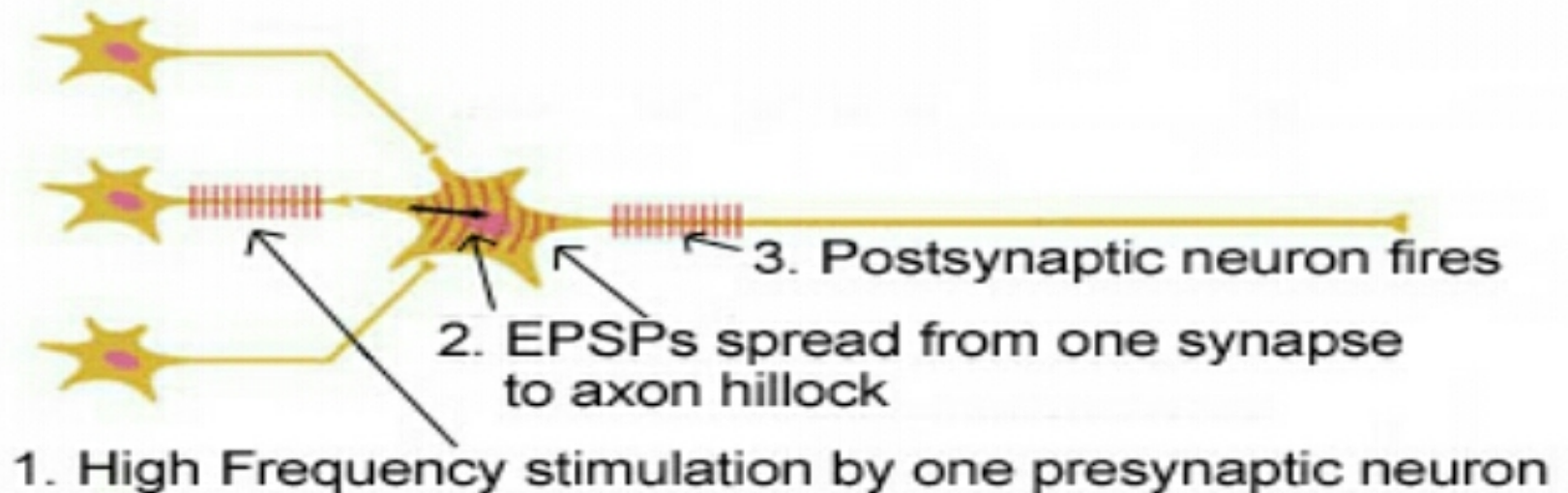
Divergence



5-Summation



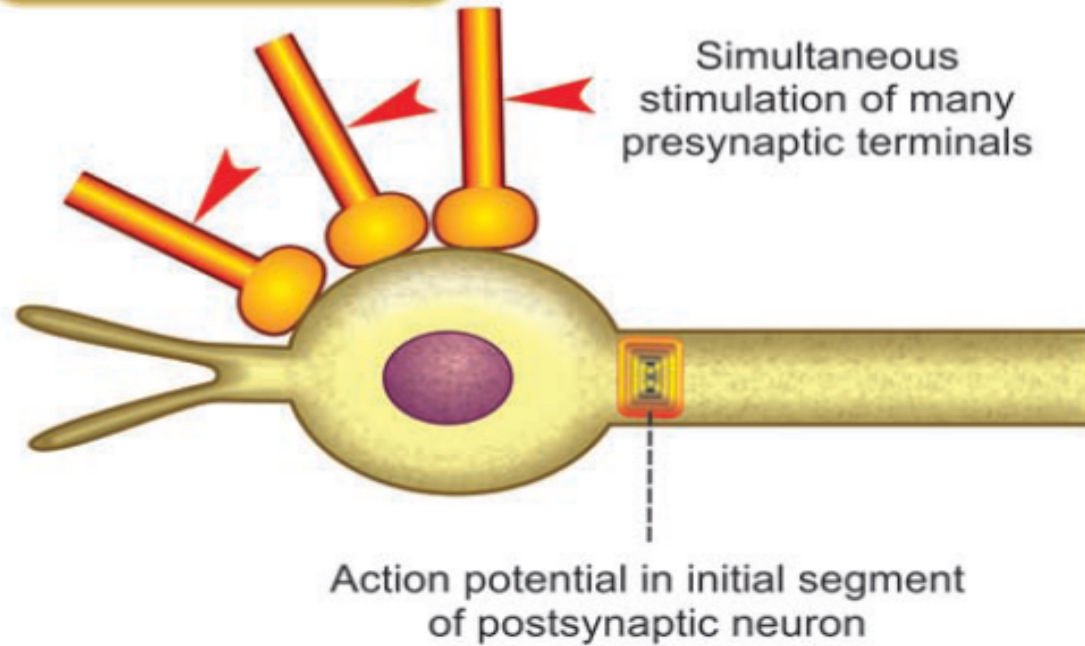
Spatial summation



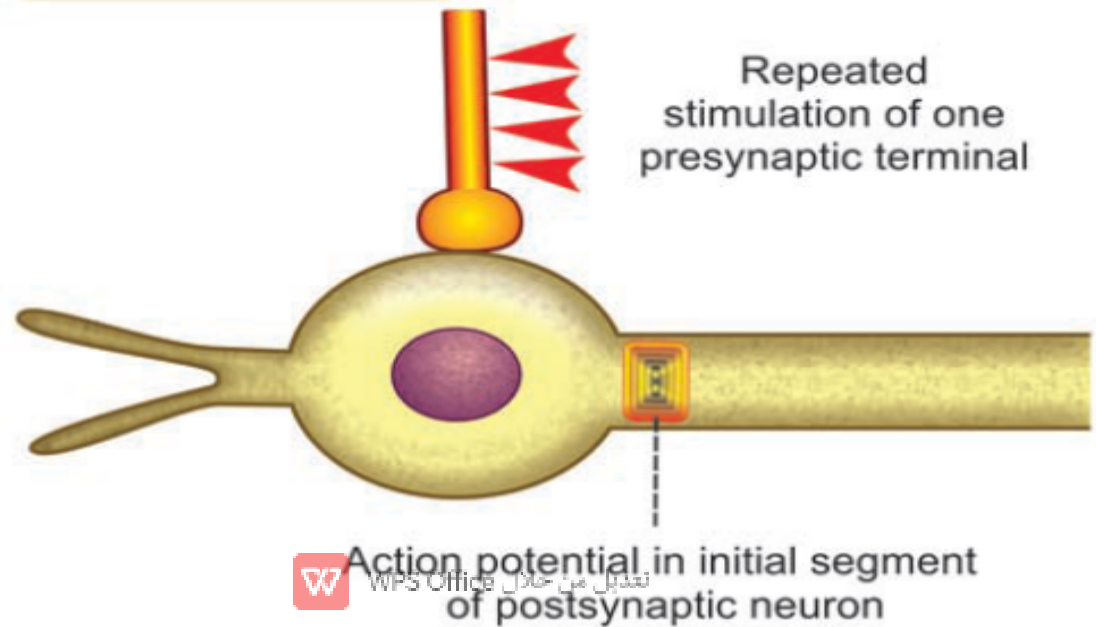
Temporal summation



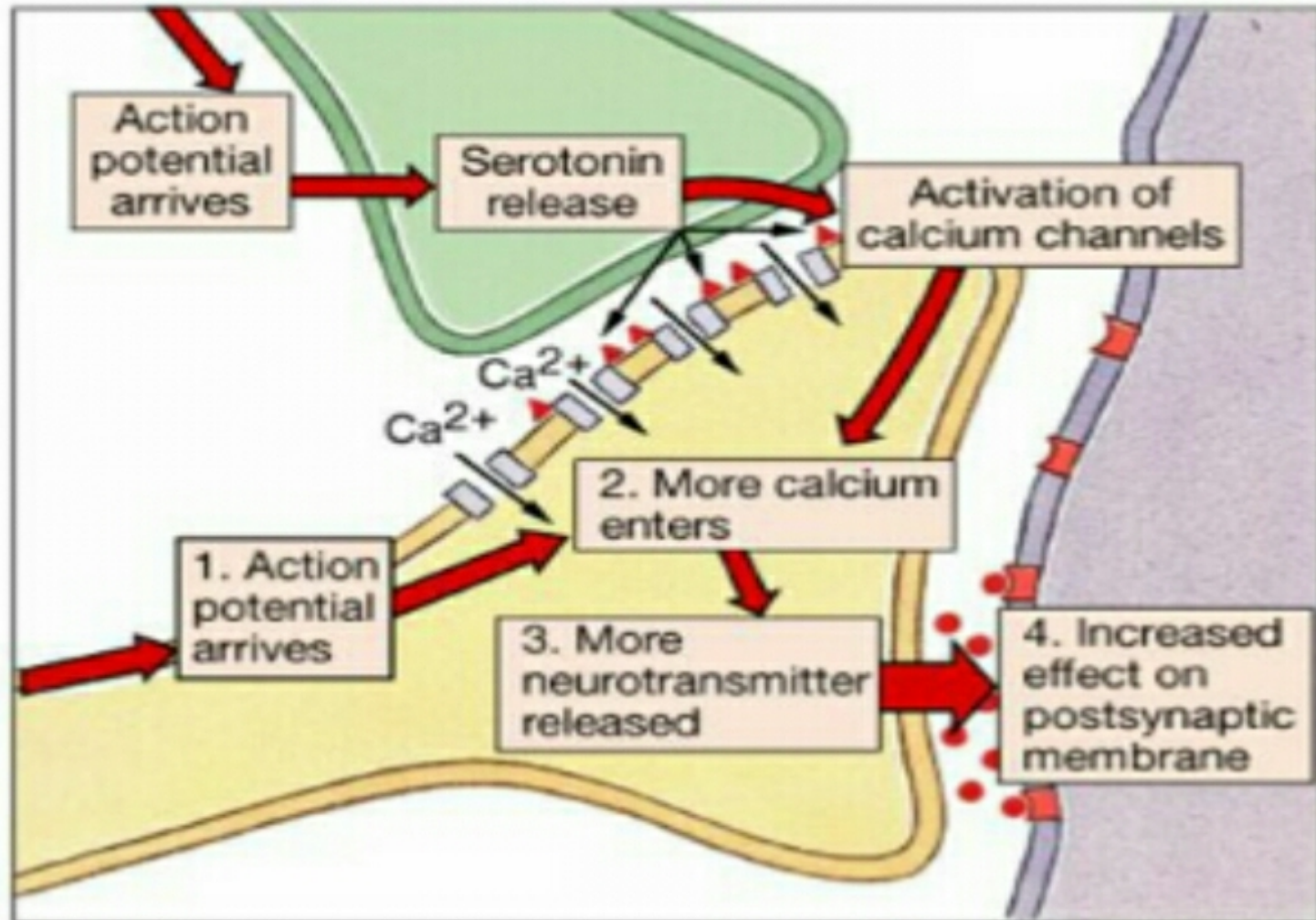
Spatial summation



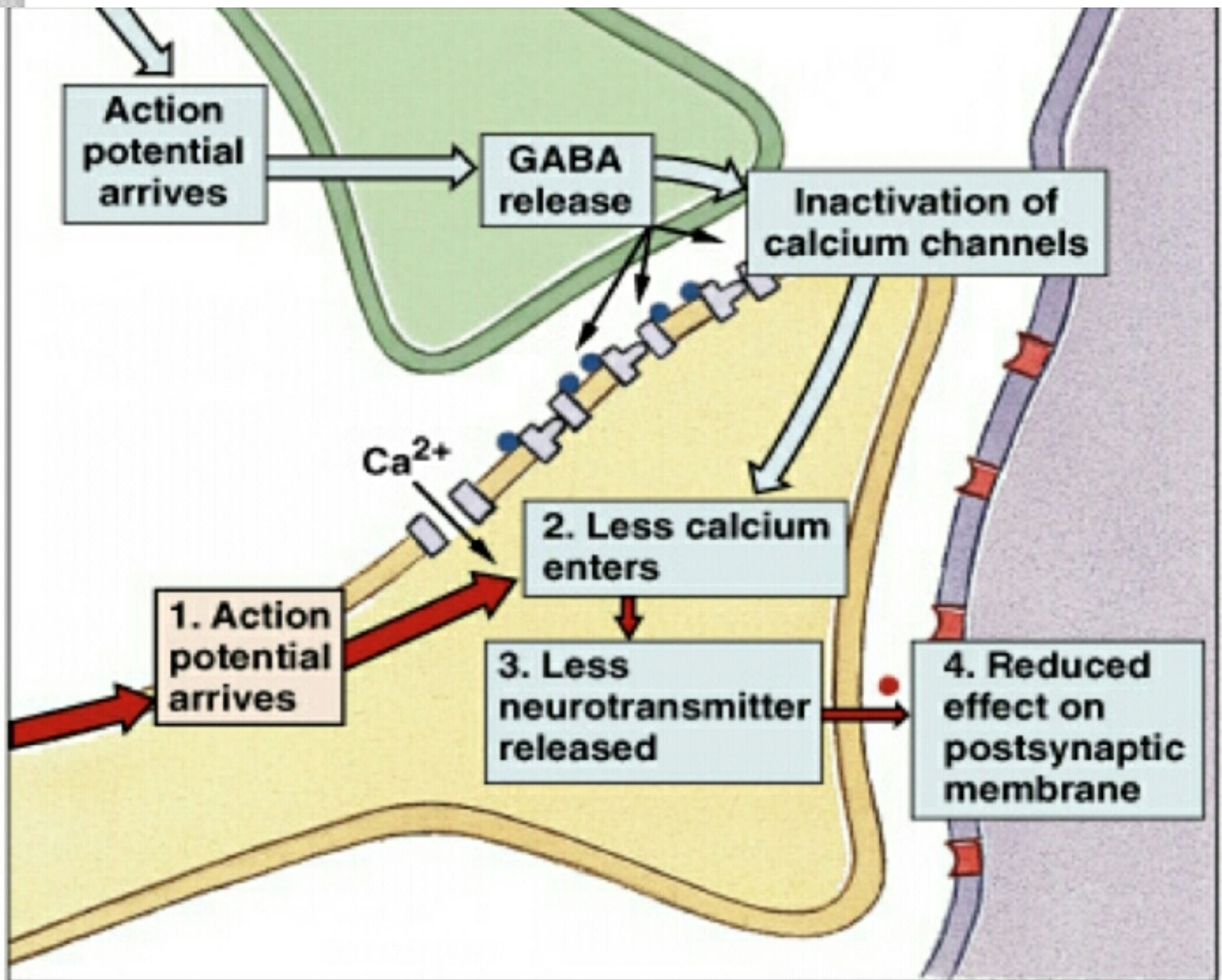
Temporal summation



6-Presynaptic Facilitation & inhibition



(b) Presynaptic facilitation

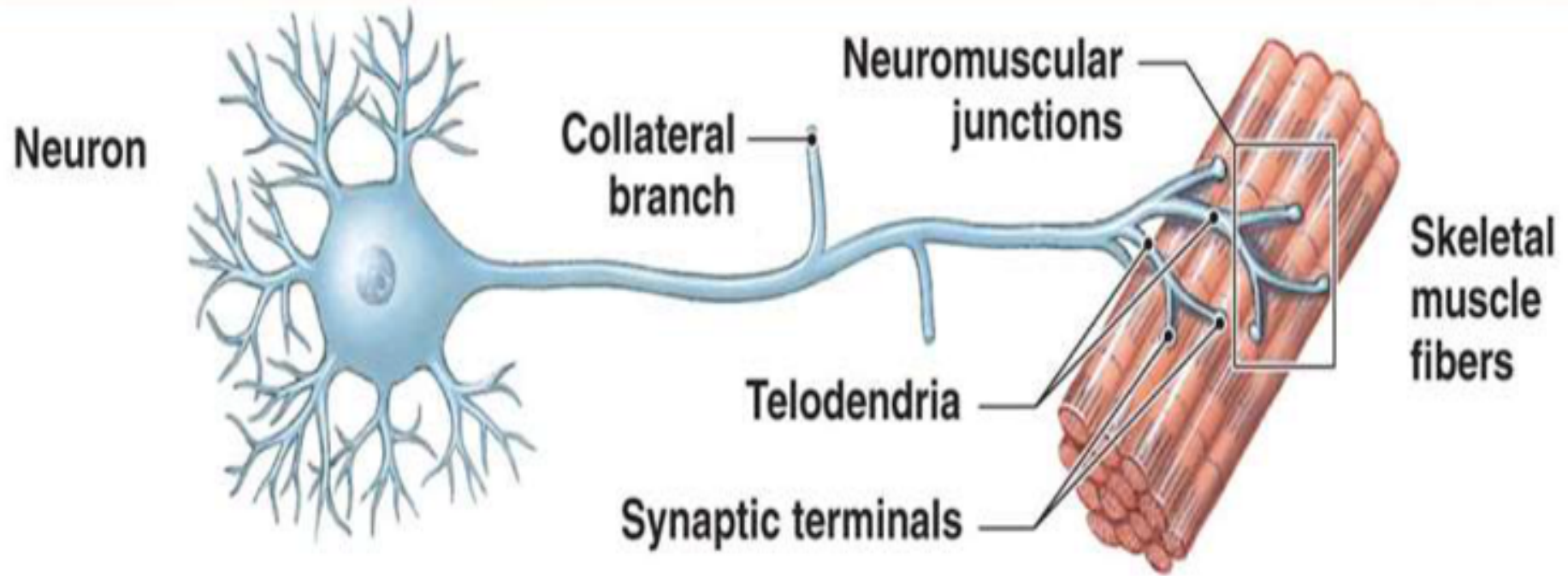


Neuromuscular junction

Definitions:

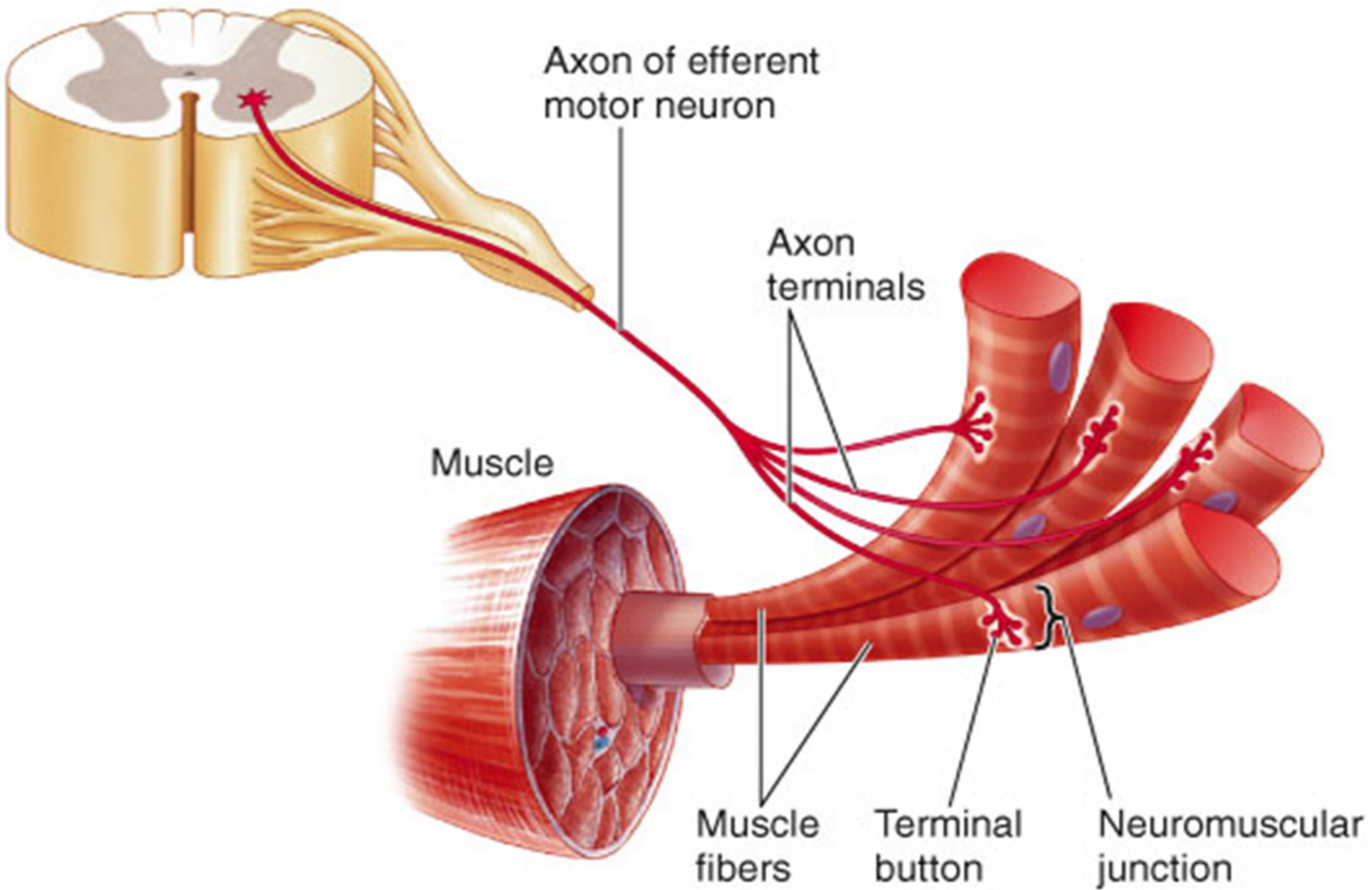
- Neuromuscular junction : the synapse between motor neuron and muscle fiber
- Motor neurons : are the nerves that innervate (power) muscle fibers
- Motor unit : single motor neuron and the muscle fiber it innervates





- Action potential from nerve is transmitted to muscle through this junction.

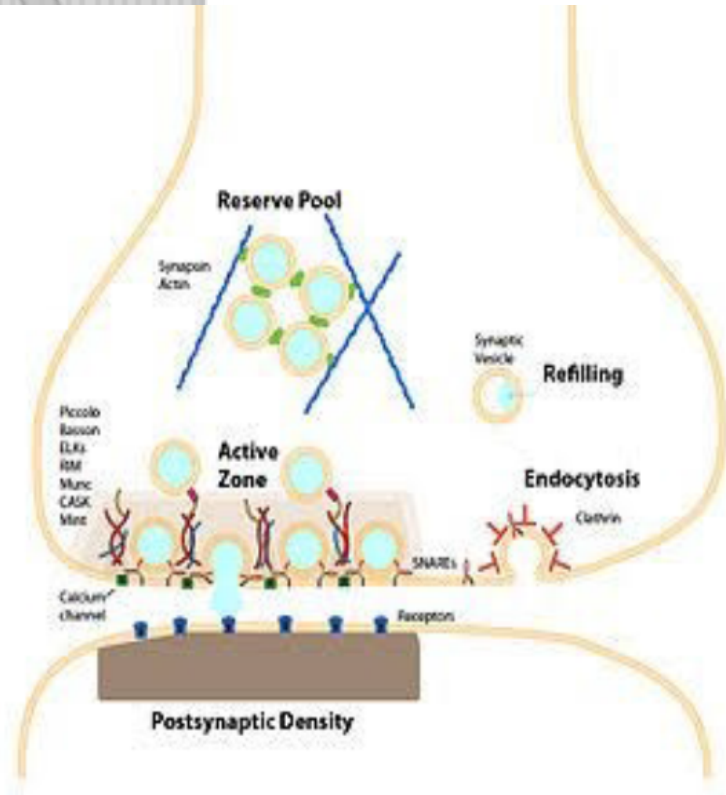
Spinal cord (section)



Structure of neuromuscular junction

1-Terminal button :

- Neuron that innervating skeletal muscle fibre known as Motor Neurons
- The terminal of each motor neuron is expanded at its tip to form Synaptic Knob (Terminal Button)
- Terminal button Contains molecules of Acetylcholine (ACh)



2-Presynaptic membrane

- The membrane of the presynapse that contain an active zone
- ** active zone is the area of attachment of the vesicles to release it's content

3-Synaptic cleft:

- It contains AchE (acetylcholine-esterase)
It hydrolyses Ach into Acetate & Choline.



4-Postsynaptic membrane:

- It's a part of sarcolemma & lies under terminal button.
- It is thrown into several folds – Junctional Folds so increases end plate membrane surface area.
- Contains acetylcholine receptors



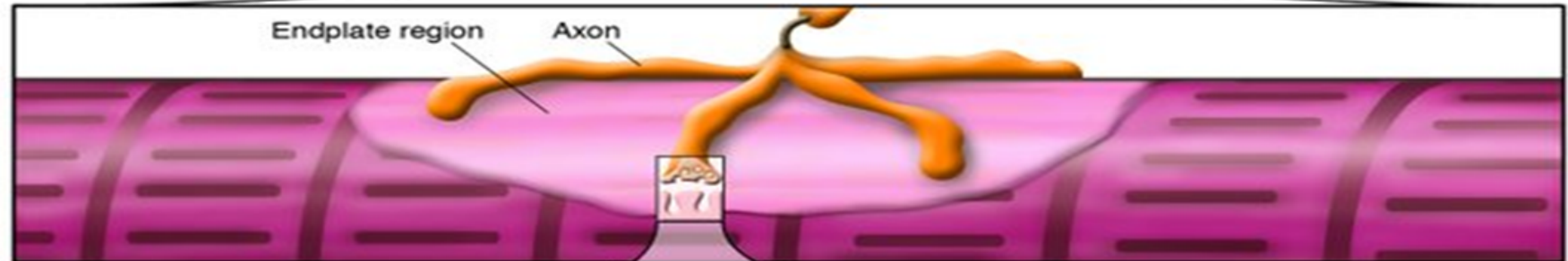
Cholinergic motor neuron
in the ventral horns of spinal
cord and brain stem

Neuromuscular
junction

Muscle
fibers

Endplate region

Axon



Presynaptic
membrane

Synaptic cleft

Postsynaptic
membrane

Junctional fold

Synaptic
vesicles

MuSK

RATL

Agrin

MASC

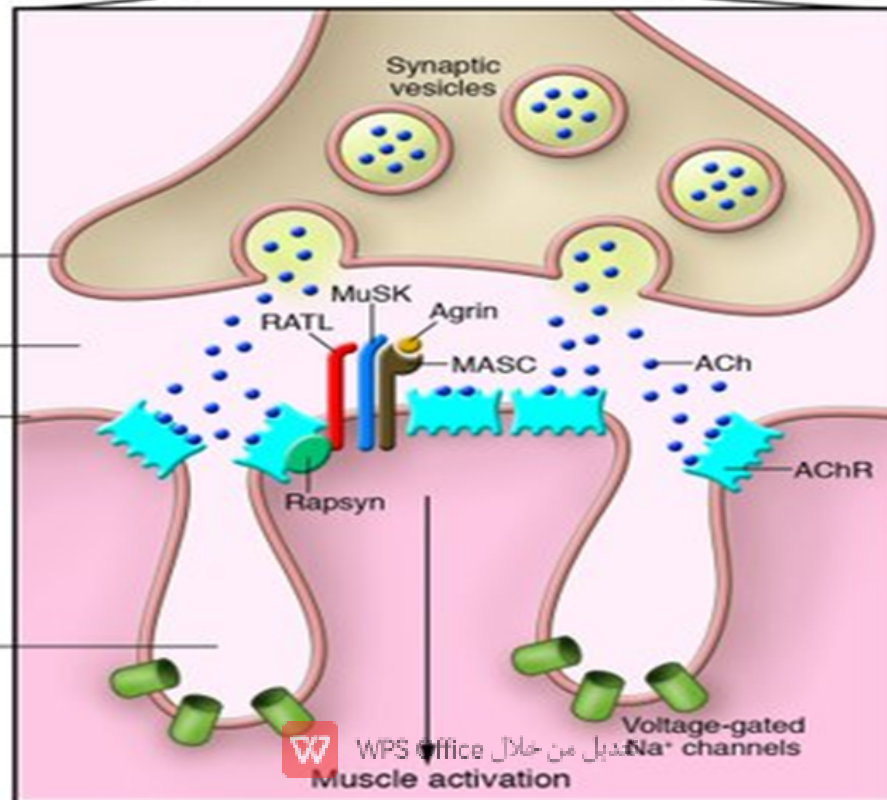
ACh

AChR

Rapsyn

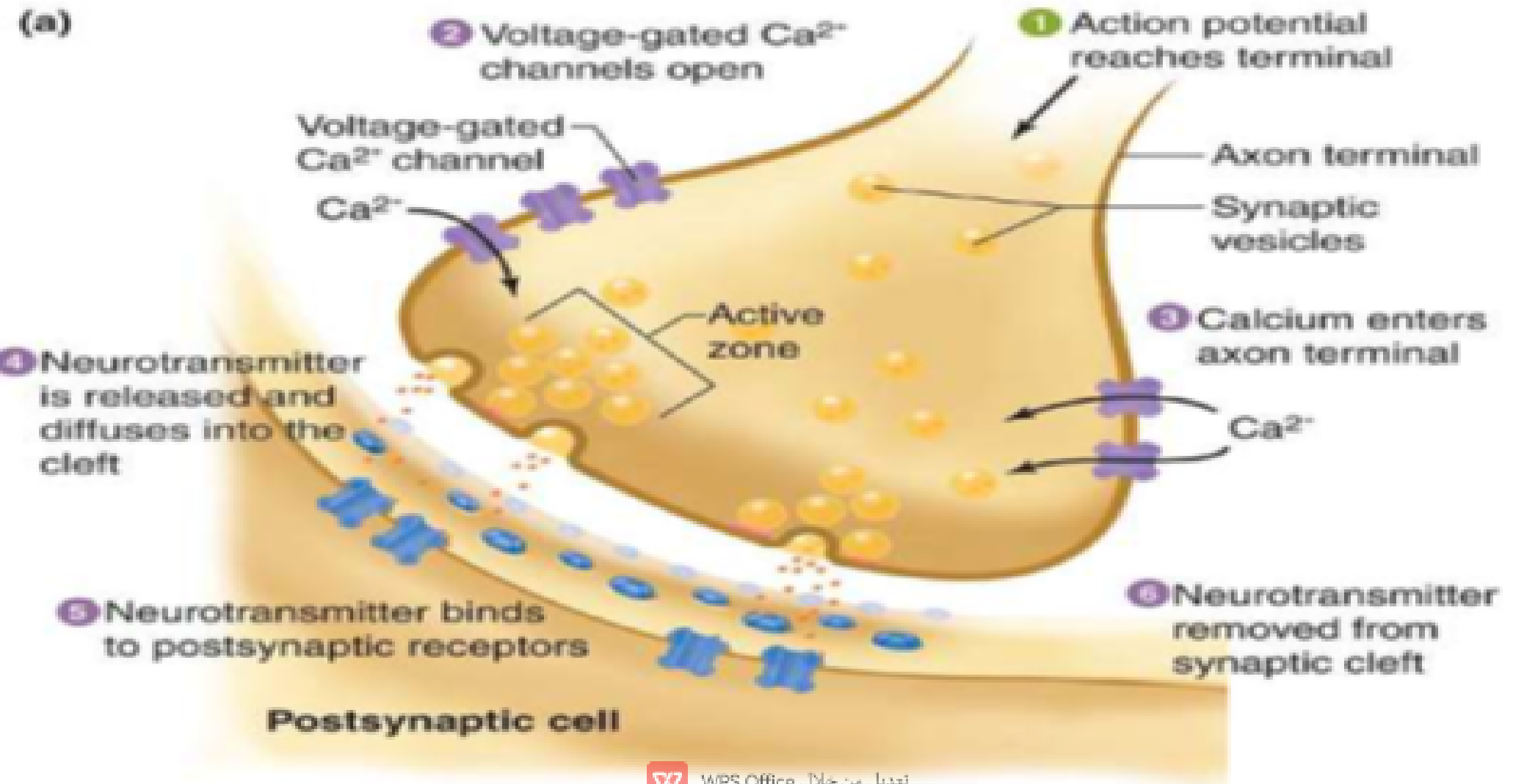
Voltage-gated
 Na^+ channels

Muscle activation

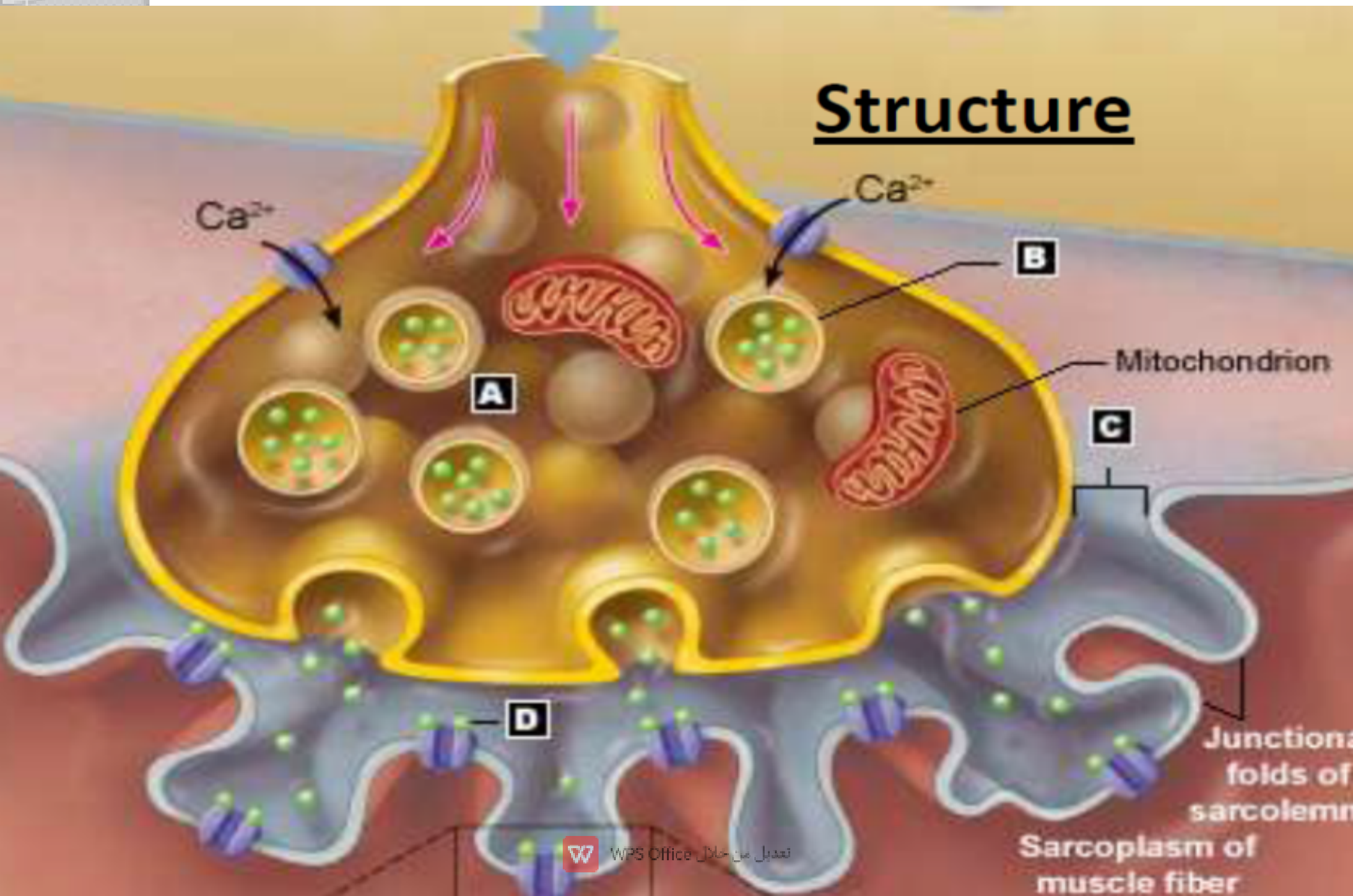


Neuromuscular transmission

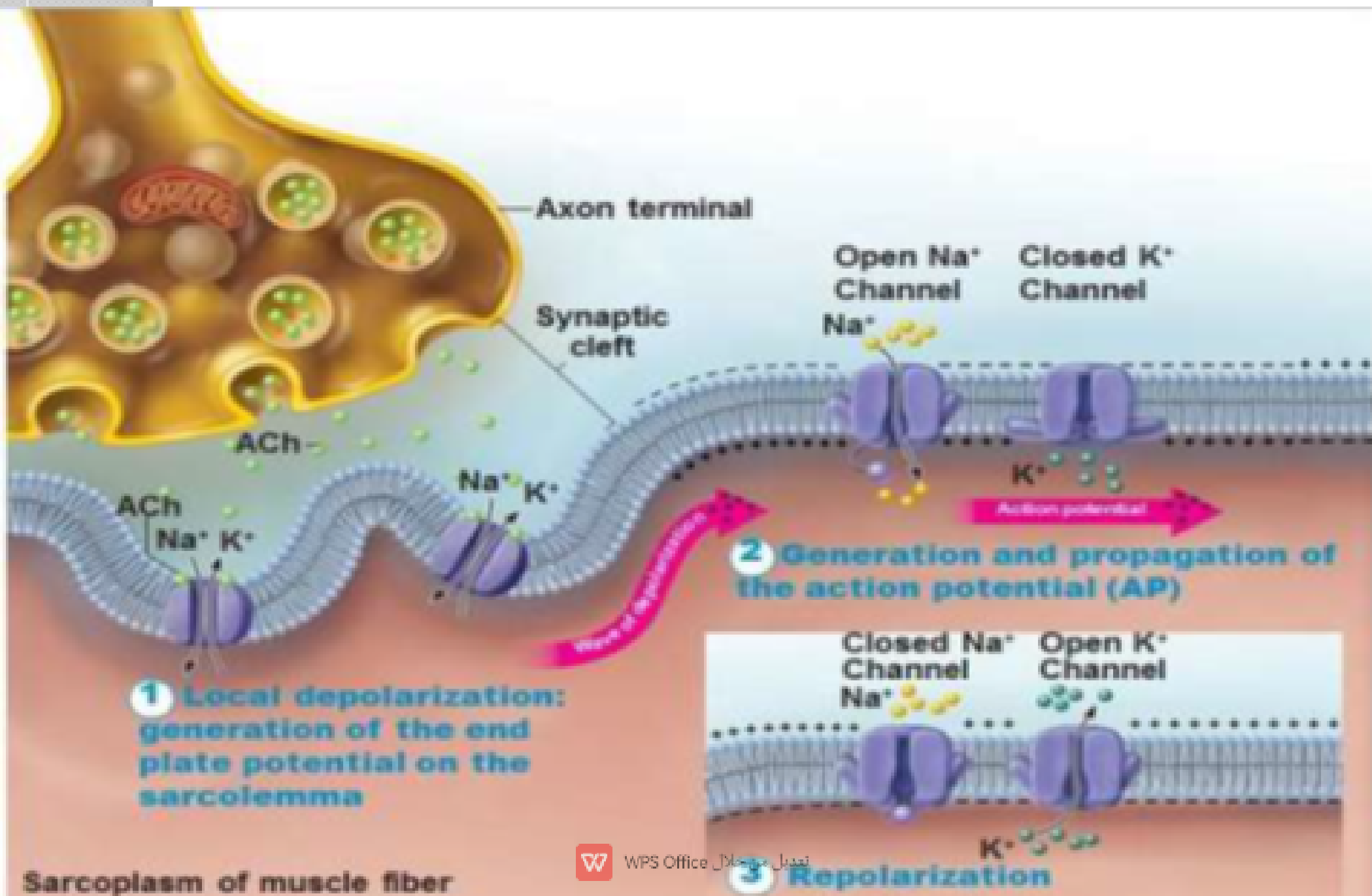
Release of acetyl choline



Action of acetyl choline



Development of end plate potential



Fate of acetylcholine

- Destruction by the enzyme acetylcholinesterase
- Reuptake
- Released in to blood stream



Neuromuscular blockers

- Curare –

blocks ach receptors by binding

- Bungarotoxin –

(snake venom extract)

blocks ach receptors

- Botulinum toxin –

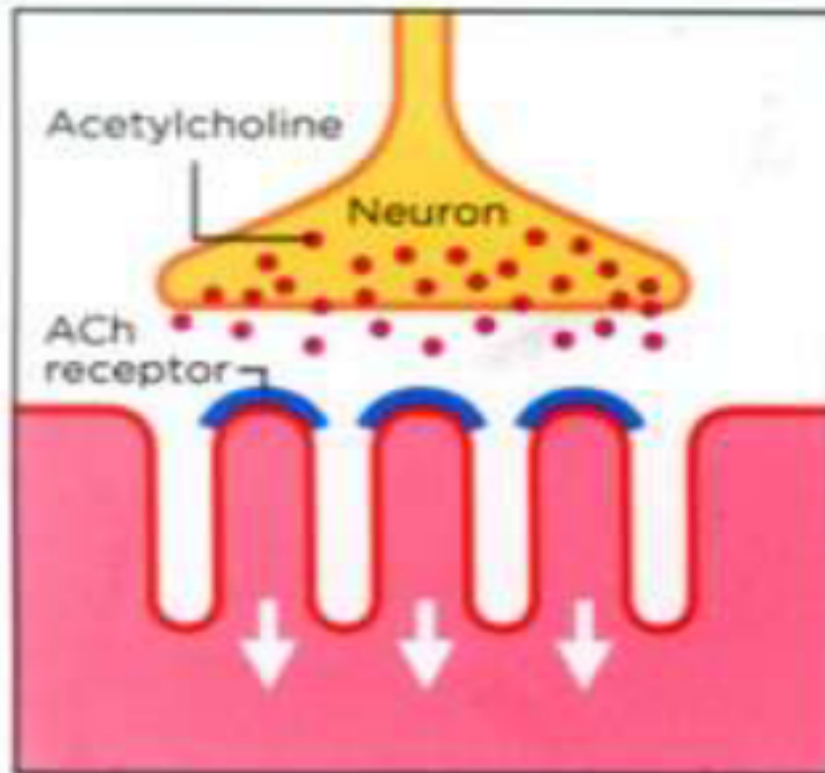
(clostridium botulinum)

prevents realease of ach from axon terminal

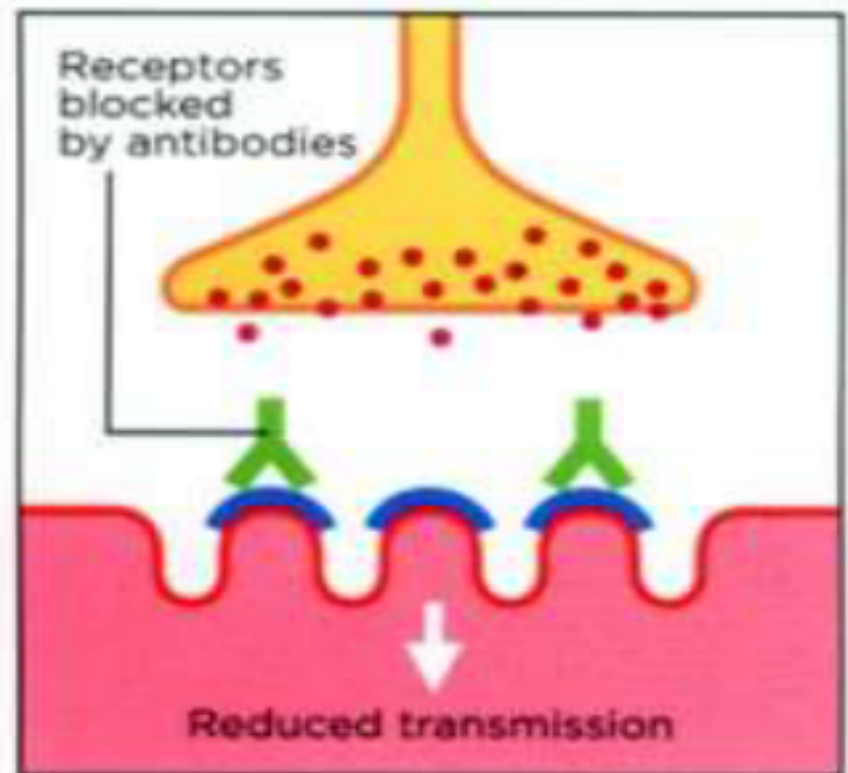


Myasthenia Gravis

- Auto immune disease
- Hereditary
- Causes muscle weakness



Normal neuromuscular junction



Neuromuscular junction in myasthenia gravis

